Wachirawit Kaewdang

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SUMMARY

Computer Science graduate (Software Engineering track) from Thammasat University. I work across machine learning for time series, with a particular focus on model explainability rather than accuracy alone, and full-stack web development in React and Node.js. Seeking Machine Learning Engineer or Software Engineer roles.

EDUCATION

Thammasat University 2022 – May 2026

BSc Computer Science (Computer and Information Science major), Software Engineering track

PROJECTS

Stock Price Direction Prediction – LSTM + Self-Attention with Explainable AI Senior project (CS403)

github.com/wachirawit-kaewdang/Project-CS403

- Built a deep learning pipeline predicting 3-day price direction for 10 US stocks across 5 sectors (2020–2025), from 27 engineered features: momentum and volatility, moving averages, RSI / MACD / Stochastic, volume and OBV, S&P 500 market context and bond yield.
- Architecture: LSTM with self-attention (3 layers, hidden size 128) over a 60-day window, trained with Focal Loss and class weights to handle imbalance.
- Compared against Random Forest, Logistic Regression, a soft-voting ensemble and a VIX market-fear feature set.
- Applied SHAP (KernelExplainer) and compared it with Pearson correlation: rankings differed by roughly 5–12 positions per feature. SHAP ranked bond yield top-5 for 6 of 10 stocks (#1 for JPM and AMZN); correlation agreed for JPM but split for others (TSLA: 2nd by SHAP, 15th by correlation).
- Evaluated results against a naive baseline: on 7 of 10 stocks the tuned model collapsed to always predicting “up”, so its F1 matched an always-up model; the ensemble’s threshold-free ROC-AUC averaged 0.568, showing short-term direction is close to random. Identified threshold and early-stopping leakage and documented fixes.
- Evaluated quarterly fundamental data (P/E, EPS, market cap) and deliberately excluded it from training after finding that constant fill-forward injected noise into the daily series; retained as descriptive analysis and documented as a limitation.
- My contribution: data collection and preparation, and design of the training approach including model architecture. Two-person project.

Python · PyTorch · SHAP · scikit-learn · pandas · NumPy · Jupyter

Recipe Web Application – Full-Stack CS369, group of 5

- React + Vite frontend with client-side routing, connected to a Node.js / Express REST API.
- My contribution: the MyRecipe management feature, recipe deletion, and integration of the frontend with the backend API.

React · Vite · JavaScript · Node.js · Express · REST APIs

HTTP Web Server from Scratch CS322 Computer Networks

- Implemented an HTTP/1.0 server in Python using raw socket programming – no frameworks – serving static HTML and image content with correct status codes and content-type headers.

Python · sockets · TCP/IP · HTTP

TECHNICAL SKILLS

Languages	Python, JavaScript, Java, C, C++, SQL
ML & Data	PyTorch, scikit-learn, SHAP, pandas, NumPy, Jupyter – time series, feature engineering, explainable AI
Web	React, Vite, Node.js, Express, Django, REST API design, HTML/CSS
Databases	MySQL, MongoDB
Tools	Git, Docker, automated testing (TDD), object-oriented design

LANGUAGES

Thai	Native
English	Professional working proficiency – technical reading and writing